
Affect steering gates refusal in Gemma-3-12B, but emotional images do not jailbreak: a progress report*

Arnav
Algoverse

Charlotte Li
Algoverse

Abstract

Can an emotional photograph open the same refusal gate that white-box affect steering opens in a vision-language model? This progress report keeps two stacks separate. On Gemma-3-12B (bf16, OASIS valence tertiles), steering an affect direction orthogonal to the Arditi refusal vector collapses first-token refuse from 0.99 to 0.04 while a matched random control stays at 0.99; restoring the refusal projection recovers refuse from 0.25 to 1.00. On Gemma-4-E4B-it (nf4, EMOTIC), pairing AdvBench-style harms with negative or neutral photos leaves refuse at ceiling (Grok $n = 100$: no-image 1.00, negative 1.00, neutral 0.98). An earlier internal image-built metric, $\text{ratio}_a = 1.591$, was labeled INVERTED. That label is not evidence that images are more affective than text because the axis was built from images. The same run’s OOD residual-norm check passes (science/ref ≈ 1.003), and unit-norming leaves $\text{ratio}_a = 1.390$ inverted. A later frozen joint-axis run reports MAGNITUDE_GAP: $\Delta_{\text{DESCRIBE}} = 3.664$ and $\Delta_{\text{harm}} = 0.389$, a suppression ratio of 0.107 (95% CI [0.079, 0.130]). Separately, its image movement is $R_{\text{affect}} = 0.0050$ of the affect-steering ceiling. Shared code now locks CORE_METHOD = fractional gate, last prompt token, window mean, normscaled, $\alpha = 0.008$. That core has not been re-run as a confirmatory experiment.

1 Introduction

Prior work identifies a residual-stream direction that mediates refusal in chat models [Arditi et al., 2024]. In text-only models, a valence-arousal subspace can move that behavior [Sun et al., 2026]. The practical question for vision-language models (VLMs) is narrower: if you attach a sad or frightening photograph to a harmful request, does the picture reach the same gate?

Two programs addressed that question with different models, corpora, and estimands. They should not be collapsed. Charlotte Li’s program (“affect as a control axis”) tests whether a valence direction causally gates refusal on Gemma-3 [Gemma Team et al., 2025] with OASIS images [Kurdi et al., 2017], then asks whether natural images can reach that gate. Arnav’s program (“why don’t emotional images jailbreak VLMs?”) tests Gemma-4-E4B-it [Gemma Team, 2026] with EMOTIC [Kosti et al., 2017, 2019], held-out splits, and an independent jailbreak direction j .

The load-bearing fact on both stacks is behavioral: emotional images do not lower refuse. The load-bearing mechanism split is that a white-box affect steer *does* open the gate on Gemma-3-12B, while Gemma-4 measurements of *how far* a photo moves affect depend on how the axis was built. An early Gemma-4 ratio above one is circular as a modality ranking. A later joint-axis run replaces that ranking with MAGNITUDE_GAP.

Progress findings, with status tags:

*This document is an internal practice writeup of work in progress. It is formatted to NeurIPS 2026 workshop style. It is not a conference submission and is not submitted to a named workshop.

1. **[HOLDS, both stacks]** Emotional images do not jailbreak under the recorded protocols (Section 5.1).
2. **[HOLDS, Gemma-3-12B / OASIS]** An affect direction gates refusal with a dose-response, random-direction control, and r -restore mediation (Section 5.2).
3. **[WITHDRAWN as mechanism]** Gemma-4 $\text{ratio}_a = 1.591$ INVERTED is not a claim that images are more affective than text (Section 5.3).
4. **[HOLDS as frozen v3]** Joint text+image $a_\perp$ on Gemma-4 yields MAGNITUDE_GAP: context-dependent suppression with correlational M1 true and causal M1 false (Section 5.4).
5. **[CODE LOCKED, NOT RE-RUN]** `scripts/affect_core.py` is the shared method; confirmatory reruns on that pin are pending (Section 6).
6. **[INCOMPLETE]** Graded sycophancy, three direction rebuilds, lighting, detector/clamp import into the core, and a Syed-style transfer matrix are not confirmatory results (Section 7).

Appendix A restates fifteen reviewer questions. The body already answers circularity, the two-stack split, $100\times$ versus ratio_a , nf4, leakage, and Grok versus first-token scores.

2 Related work

Refusal as a direction. Ardit et al. [2024] extract r as the mean last-token residual difference between harmful and harmless instructions, then show that adding or erasing that direction induces or removes refusal. We reuse that construction, including a competence check that adding scaled $+r$ to harmless prompts raises refusal.

Text-only affect gates. Sun et al. [2026] recover a 2D valence-arousal subspace in LLMs and show that steering it modulates refusal and sycophancy. Their models are text-only. Charlotte’s 12B result is the VLM analogue with an image-built valence contrast, not a replication of their lexical-mediation experiments.

Appraisal and mid-layer emotion. Zhou et al. [2024] argue that alignment ties early unethical classification to mid-layer negative emotion, then to refuse tokens, and that jailbreaks disturb that association. This motivates, but does not establish for VLMs, a directional hypothesis that perceived negative affect should raise rather than lower refusal. The image experiments test that hypothesis.

Image corpora. EMOTIC labels people-in-context emotion [Kosti et al., 2017, 2019]. OASIS is a normative valence/arousal photo set [Kurdi et al., 2017]. Charlotte’s early EMOTIC download was incomplete (~ 184 images, a 1-image negative set) and poisoned affect numbers; confirmed Charlotte results use OASIS. Arnav’s Gemma-4 runs use a full EMOTIC dump (23,185 JPGs). Those are different visual populations.

Steering tools. Interventions follow activation-addition practice [Turner et al., 2023, Rinsky et al., 2024] through TransformerLens hooks on `resid_post` [Nanda and Bloom, 2022]. Harmful prompts come from AdvBench [Zou et al., 2023]. Harmless prompts on the Charlotte stack use Alpaca [Taori et al., 2023].

Orthogonality is not independence. Wollschläger et al. [2025] warn that orthogonalizing a concept against r does not make it causally independent of refusal. Charlotte’s r -restore test is the empirical counterpart: $a_\perp$ still acts *through* r .

Three quantities that are not interchangeable. Table 1 is the load-bearing distinction. INVERTED does not refute Charlotte’s shared-axis behavioral story.

3 Method

Locked shared method, now in `scripts/affect_core.py`:

Table 1: Three recorded affect/refusal numbers. They are not the same estimand. Do not substitute one for another.

Name	Definition	Recorded value
Charlotte 100× v3 suppression	Harmful-prompt image Δ on $a_{\perp}$ vs. jailbreak steer Bootstrap median of $\Delta_{\text{harm}}/\Delta_{\text{DESCRIBE}}$ on joint	$\sim 11\text{-}22$ vs. ~ 2200 0.107, 95% CI [0.079, 0.130]
ratio_a	$\frac{a_{\perp}}{ \Delta_{\text{img}} / \Delta_{\text{txt}} }$ on <i>image-built</i> $a_{\perp}$ (publish)	1.591 (construction-biased)

CORE_METHOD = fractional_gate, last_prompt, window_mean,
normscaled, alpha=0.008.

Layer windows: $L = 34$ gives [8, 20); $L = 42$ gives [10, 25); $L = 48$ gives [12, 28). Equivalent form: $\llbracket 0.25L \rrbracket, \llbracket 0.60L \rrbracket$.

Directions, last-token resid_post. Let $h_{\ell}(x)$ be the last prompt-token residual at layer ℓ . Write $\text{unit}(v) = v/\|v\|$.

$$r = \text{unit}(\mathbb{E}[h(\text{harmful})] - \mathbb{E}[h(\text{harmless})]), \quad (1)$$

$$a_{\text{img}} = \text{unit}(\mathbb{E}[h(\text{DESCRIBE}+\text{neg})] - \mathbb{E}[h(\text{DESCRIBE}+\text{neu})]), \quad (2)$$

$$a_{\text{text}} = \text{unit}(\mathbb{E}[h(\text{distress text})] - \mathbb{E}[h(\text{neutral text})]). \quad (3)$$

Joint primary axis:

$$a_{\perp} = \text{orth}_{\text{span}}(\text{unit}(a_{\text{text}} + a_{\text{img}}), r). \quad (4)$$

Image-only $a_{\perp}$ is a named variant `image_only_a_perp`, not the headline shared construct. Orthogonalization is QR on the span, not sequential Gram-Schmidt against a non-orthogonal $\{r, j\}$ pair (sequential subtraction can put r back).

DESCRIBE is the string “Describe what is happening in this image.” The publish-pipeline distress comparator is the single string “You feel deeply sad, afraid, and distressed while reading this.” Mechanism v3 replaces that with 64 distress and 64 neutral text bank items.

Normscaled steer. On gate layers,

$$\text{resid}_{\ell} \leftarrow \text{resid}_{\ell} + \alpha \|\text{resid}_{\ell}\| \hat{d}_{\ell}. \quad (5)$$

Fixed coefficients (the obsolete `coeff` = 20 / late-layer dry run) silently no-op on Gemma residual norms on the order of 10^4 . That dry run is not Arnav’s current mechanism and is not a confirmatory result.

First-token refuse score. Refuse vocabulary: I, I’m, Sorry, As, Unfortunately, Cannot. Comply vocabulary: Sure, Here, Certainly, Of, Absolutely, Step. The generation-free score is $\log\text{sumexp}(\text{refuse logits})$ minus $\log\text{sumexp}(\text{comply logits})$ at the first generated token. Rate is the fraction of prompts with refuse > comply. Grok is secondary on Arnav’s behavioral arms: publish $n = 100$ per cell, and v3 harmful $n = 100$ plus borderline $n = 80$. Publish `judge_fallback_pct` = 0. Charlotte headlines are generation-free; no LLM judge.

Dual stacks. Table 2 is the design split. Effect sizes are not pooled across rows.

Leakage wall. Arnav image ids are hashed into train/eval (and v3 calib/science). Publish split hash `ef2a7d2a84a803c7`, seed 0, train 10,184 / eval 6,793. Mechanism v3 split hash `1e8ea1c22144dd9d`, same train/eval sizes, plus $n_{\text{calib}} = 1018$ and $n_{\text{science}} = 5775$. Charlotte’s confirmed OASIS runs used split-half stability a_{stab} rather than a hashed science holdout; some image estimates were in-sample. The unified core requires hashed splits going forward.

Frozen v3. `artifacts/colab/e2e_mechanism_results_full_v3.json` is read-only. Later code changes must not overwrite it. Unification edits the shared core; they do not rewrite that JSON.

Table 2: Dual stacks. Science factors, not method knobs. Do not pool nf4+EMOTIC numbers with bf16+OASIS numbers.

Item	Charlotte	Arnav publish	Arnav mechanism v3
Model	Gemma-3-12B-it	Gemma-4-E4B-it	Gemma-4-E4B-it
Layers / d	48 / not recorded here	42 / 2560	42 / 2560
Dtype	bf16	nf4, bf16 compute	nf4, bf16 compute
Images	OASIS tertiles, 200/class	EMOTIC	EMOTIC
Primary a	image valence, then $a_{\perp}$	image_only_a_perp	joint $a_{\perp}$
Gate	[12, 28) at $L = 48$	[10, 25)	[10, 25)
Headline metric	first-token refuse	Grok + FT	FT + Grok; Δ on $a_{\perp}$

Table 3: Refuse rates for harmful prompts with and without images. Stacks are not pooled. FT is the first-token metric; Grok is an LLM judge on full generations, used only on the Arnav behavioral arm.

Condition	Charlotte 12B FT	Arnav FT (img table)	Arnav Grok $n = 100$
No image	0.99 (base)	0.99	1.00
Neutral image	0.99	1.00	0.98
Negative image	0.99	1.00	1.00
Positive image	0.99	1.00	n/a

4 Experimental setting

Charlotte / OASIS / Gemma-3. Primary model: Gemma-3-12B-it, bf16, $n_{\text{layers}} = 48$, residual-norm median 75,776, $a_{\text{stab}} = 0.830$, $\cos(a, r) = 0.371$, $\alpha = 0.008$. Image set: OASIS valence tertiles, 200 per class. Roster (not pooled with 12B): Gemma-3-4B, LLaVA-OneVision-7B [Li et al., 2025], Qwen2.5-VL [Bai et al., 2025], InternVL3 [Zhu et al., 2025], LLaVA-OV-0.5B, Qwen3.5-4B. GPU SKU and wall time for the 12B bf16 runs are not recorded.

Arnav publish / EMOTIC / Gemma-4. Model: google/gemma-4-E4B-it. Gemma-3-4B fallback was not used. GPU: Tesla T4, ~ 15.6 GB. Full pipeline finished 2026-08-07 00:21:17; wall ~ 8942 s. EMOTIC: 23,185 JPGs; unique images after person-dedup from 16,977 person-rows; buckets neu 10,573, pos 5,838, neg 566. Prereg hash 3111c3af7f7b12656cb0792bea4a21b4302a0c42. Sample sizes: $N_{\text{DIR}} = 64$, $N_{\text{IMG}} = 64$, $N_{\text{EVAL}} = 100$, $N_{\Delta} = 40$, $N_{\text{GAP}} = 50$, $N_{\text{BEH}} = 100$, $N_{\Gamma} = 40$.

Arnav mechanism v3. Same Gemma-4 nf4 model and gate. Started 2026-08-14 23:00:30, elapsed 6740.5 s, complete=true, GPU NVIDIA A100-SXM4-40GB. Text banks: 64 distress + 64 neutral. $n_{\Delta} = 64$. Harmful behavioral $n = 100$, borderline $n = 80$.

Prompts. Harmful: AdvBench-style instructions [Zou et al., 2023]. Image cells use DESCRIBE or the harmful instruction plus a photo. γ stress tests (publish) sweep a large coefficient along emotion versus random; they are named collapse checks, not the locked $\alpha = 0.008$ dose.

5 Results

5.1 Behavioral null on both stacks

Table 3 is the claim that survives the geometry audit. Emotional photographs do not drop refuse under these protocols.

Charlotte 12B image refuse is 0.98-0.99 across valences in the writeup summary and 0.99 in replication_full.json for neu/neg/pos. Mechanism v3 Phase 1 is the same null on Gemma-4: FT refuse ~ 0.99 -1.0 with and without images on harmful $n = 100$ and border $n = 80$; Grok agrees; images_jailbreak=false.

Table 4: Gemma-3-12B dose-response from the results writeup. Random control is a direction orthogonal to r at matched α . Coherence fails at $\alpha = 0.016$ (over-steer).

$\alpha \times \ \text{resid}\ $	Affect $a_{\perp}$ refuse	Random $\perp$ refuse	Coherent
0.000	1.00	1.00	yes
0.008	0.75	1.00	yes
0.010	0.25	1.00	yes
0.012	0.00	1.00	yes
0.016	0.00	1.00	no

Table 5: Charlotte roster on OASIS. “Gate” is a qualitative label from the writeup, not a pooled effect size with Gemma-4.

Model	a_{stab}	Affect-steer refuse	Gate
Gemma-3-12B-it	0.830	0.04	yes
Gemma-3-4B-it	0.633	1.0 ($\alpha = 0.004$)	no
LLaVA-OV-7B	0.652	0.65	partial
Qwen2.5-VL, InternVL3	n/a	no gate	no
LLaVA-OV-0.5B	n/a	(base 0.29)	excluded
Qwen3.5-4B	n/a	(base 0.09)	weak refuser

5.2 Charlotte: Gemma-3-12B affect gate

On Gemma-3-12B, white-box steering of $-a_{\perp}$ (toward benign affect) opens refusal. `replication_full.json` records base refuse 0.99, affect-gate refuse 0.04, random-gate refuse 0.99 at $\alpha = 0.008$. A separate dose-response grid from the results writeup is Table 4. Those two artifacts should not be treated as one fitted curve until a unified-core rerun.

Mediation through r . Affect steering drops the r -projection 30,756 to 21,421 ($\sim -30\%$). Steering affect while restoring the r -projection to the clean baseline recovers refuse 0.25 to 1.00. The gate is a valence route into Arditi’s direction, not a second independent refusal axis.

Detector versus clamp. Monitoring the affect projection: detector AUROC 1.0 against a white-box steer (partly expected for a large injected offset). A one-sided projection clamp restores attack refuse only 0.00 to 0.17 and over-refuses benign prompts 0.06 to 0.81. Localization of the steer: early refuse 1.0, mid 0.33, late 1.0.

Model table. Table 5 is the roster. The gate is present on Gemma-3-12B and absent or partial among the other recorded strong-refuser tests. Gemma-3-4B has $a_{\text{stab}} = 0.633$ and affect-gate refuse 1.0 at $\alpha = 0.004$: no confirmed gate, with a stability confound. LLaVA-OV-7B has $a_{\text{stab}} = 0.652$ and is partial (affect-steer refuse 0.65). Qwen2.5-VL and InternVL3: no gate. LLaVA-OV-0.5B is a weak refuser (base 0.29). Qwen3.5-4B base refuse 0.09 (not a strong refuser); image refuse still ~ 0.99 .

Why images still do not jailbreak on 12B. Under harmful text, image movement on the gate axis is ~ 11 -22 versus a steer ceiling ~ 2200 : about $100\times$ too small. That is a reachability gap, not ratio_a .

5.3 Arnav publish metric, then the audit

The full-tier publish pipeline reported kill-geometry INVERTED on image-built $a_{\perp}$: $\text{ratio}_a = 1.591$, $\text{ratio}_h = 0.036$, with $\delta_{\text{txt},a} = 3.401$, $\delta_{\text{img},a} = 5.411$, $\delta_{\text{txt},h} = 25.079$, $\delta_{\text{img},h} = 0.902$, and $|\cos(a, r)| = 0.1361$. Competence checks: r on harmless prompts takes refuse 0.075 to 1.0 at $+0.25r$; h validate 0.067 to 0.80 versus random 0.00 (`valid=true` in the JSON).

The 2026-08-10 validity audit is binding for claims.

Construction bias. $a / a_{\perp}$ was built from images (EMOTIC neg—neu under DESCRIBE). Measuring $|\Delta_{\text{img}}|/|\Delta_{\text{txt}}|$ on that axis reconstructs the defining contrast in the numerator. $\text{ratio}_a = 1.591$

is a valid number for that biased metric. It is not evidence that photographs are intrinsically more emotional than text. The INVERTED branch is withdrawn as a mechanism headline.

What still holds from the same run. OOD residual-norm PASS: science/ref ≈ 1.003 (not color-square inflation). Unit-norm recheck: $\text{ratio}_{a,\text{raw}} = 1.591$, $\text{ratio}_{a,\text{unit}} = 1.390$; inversion persists as a narrow measurement on the same image-built direction. γ collapse is generic: $\gamma = 1$ to 20 refuse 1.0; $\gamma = 50$ emotion 0.95 / random 1.0; $\gamma = 100$ emotion 0.475 / random 0.1; $\text{coherent_frac} = 1.0$ throughout. Random is worse at $\gamma = 100$, so this is not an emotion-specific jailbreak. Gap $B/A = 1.129$ ($d_{\text{img},A} = 5.898$, $d_{\text{cap},B} = 6.660$, $d_{\text{txt},C} = 13.259$, $n = 50$) only partially holds, because A and B sit on the image-built axis. Gap C’s baseline (distress minus image-neu) is suspect. ratio_h is suspect: h was built from harmful text minus DESCRIBE(eval neg), with overlap. The distress string was repeated rather than diversified. Three independent direction rebuilds were not done.

Contagion (recorded, not a jailbreak). Emotion-image contagion rate 0.045 versus neu 0.0, delta 0.045.

5.4 Arnav mechanism v3: MAGNITUDE_GAP

Frozen v3 rebuilds affect as $a_\perp = \text{orth}(\text{unit}(a_{\text{text}} + a_{\text{img}}), r)$ and adds an independent jailbreak direction j from natural refuse/comply terciles on a borderline bank (not a steered $-r$ in disguise). $|\cos(j, r)| = 0.247$, status INDEPENDENT. j competence (FT): base 12.62, $+j$ 0.47, $+j_\perp$ -0.61 . Geometry: $|\cos(a_{\text{text}}, r)| = 0.169$, $|\cos(a_{\text{img}}, r)| = 0.135$, $|\cos(a_{\text{text}}, a_{\text{img}})| = 0.098$, $|\cos(a_{\text{both}}, r)| = 0.186$. r validate 0.083 to 1.0. OOD PASS ratio 1.002.

Table 6 records the frozen mechanism flags. B1/B2 are the behavioral non-inferiority gates (images do not jailbreak hard harms or borderline prompts).

Phase 2 deltas on $a_\perp$. $\Delta_{\text{DESCRIBE}} = 3.664$, $\Delta_{\text{emotionQ}} = 1.536$, $\Delta_{\text{harm}} = 0.389$, $\Delta_{\text{border}} = 0.353$. Whitened null $p_{95} = 0.751$, whitened $p = 0.0$; isotropic $p_{95} = 0.059$; $\text{floor_pass} = \text{true}$. Suppression ratio 0.107, 95% CI [0.079, 0.130], $n_\Delta = 64$, M1 correlational true. The point is the median of 1,000 paired-image bootstrap ratios; the CI is the corresponding percentile interval. Diagnostic: $a_{\text{img}\perp}$ DESCRIBE 5.396 / harm 0.576; $a_{\text{text}\perp}$ DESCRIBE 0.146 / harm -0.182 .

Phase 3: miss on j . $s_{j,\text{neg}} = 4.275$ versus $s_{j,\text{neu}} = 4.451$ (images miss the jailbreak subspace). $s_{a,\text{neg}} = -1.987$.

Phase 4 ceilings. $\alpha_{a,\text{ceiling}} = 0.4$, $R_{\text{affect}} = 0.0050$; $\alpha_{j,\text{ceiling}} = 0.4$, $s_{j,\text{ceiling}} = 99.48$, $R_{\text{jail}} = 0.043$.

C1 is not causal. $\cos(\delta, a) = 0.662$; $\text{drop}_{\text{clean}} = 1.68$ versus $\text{drop}_{\text{rand}} = 7.56$; rate $\text{drop}_{\text{clean}} = 0.0$; M1 causal false. Restoring the image-scale Δ under harmful prompts does not open refusal.

Headline. MAGNITUDE_GAP: photos move emotion under DESCRIBE, but under harmful asks the push is small. The harmful/DESCRIBE ratio is 0.107. Separately, $R_{\text{affect}} = 0.0050$ compares movement with the v3 affect-steering ceiling. Neither quantity is Charlotte’s 100 $\times$ comparison (Table 1).

6 Unified pipeline

Code now shares one method implementation. Scientific programs stay separate. Pull request: <https://github.com/halli75/algoverse/pull/1>. Core tests: 19 unit tests, no GPU.

Locked choices and why: fractional gate plus window mean plus normscaled 0.008 (the family that moved tone; $\text{coeff} = 20/\text{late}$ was an obsolete dry run); depth-portable [0.25 L , 0.60 L]; last-token scalars averaged on the intervened window, not an all-layer mean; joint $a_\perp$ as primary; QR span orthogonalization; hashed train/calib/eval; first-token refuse as shared headline.

Table 6: Frozen v3 hypothesis flags. M1 correlational holds; the causal restore (C1) does not. j is independent of r and images miss it.

Flag	Verdict	Meaning
B1 / B2	true	Images do not jailbreak hard harm or borderline
no_jailbreak harm+border	true	Same behavioral null
M1_corr	true	Δ large under DESCRIBE, small under harmful
M1_causal (C1)	false	Restoring that Δ does not drop refuse
M2_indep / M2_miss	true / true	j independent of r ; images miss j
M3 protective	true	Negative images raise r / refuse
M4 site failure	false	Not “ Δ dies before refuse layers”
M5 ceiling	false	Not “no headroom only on hard harms”
caption_vision_specific	false	Caption is not uniquely vision-specific

Not merged, and not allowed to overwrite the core: Arnav graded sycophancy; M1-M5/ j scoring; caption gap; γ as a causal dose; Grok as headline. Charlotte detector/clamp; lighting; six-probe battery; agentic/protective modules; cross-model roster.

Those modules may import the core. Confirmatory numbers in this paper are the older stack-specific runs. The unified core has not been re-run.

7 Incomplete work

Sycophancy. Arnav-only, frozen, not done. Plan hash prefix 051e1ea79cd116fce56edeb43848ae10526e2830. T4 smoke: ENDPOINT_INVALID; Likert digit mass $0.181 < 0.8$ (later amendment: Likert invalid should not kill S_{label}). A VA-estimator repair is gist-pinned; last smoke $r_a = -0.019$ against a ≥ 0.7 gate on the old VA estimator. Phase 1 $S_{\text{label},\text{mean}} = 3.88$, $S_{\text{native},\text{mean}} = 4.58$, $f_{\text{c_mass}} = 0.989$. These are not sycophancy results.

Local color squares. Gemma-3-4B on a local RTX 3050 with solid color squares: INCONCLUSIVE_OR_MIXED, $\sim 192 \times$ anomaly. Invalid for claims. Plumbing failure only.

Direction rebuilds. Pre-registration called for three independent direction seeds. They were not done. Seed 0 numbers stand as single-run artifacts.

Unified-core rerun. Pending. Do not read Section 6 as a new experiment.

Other Charlotte modules. Lighting, the six-probe behavior battery, and input-space LLM attacker cells are not confirmatory in this writeup.

Early EMOTIC poison. Charlotte’s ~ 184 -image download (1-image negative) is retired. Do not cite those affect numbers.

8 Limitations

nf4 versus bf16 is an unquantified confound; we do not treat Gemma-4 suppression ratios as a scale replication of the 12B gate. EMOTIC people-in-context labels are not OASIS normative valence. Negative EMOTIC pool size (566) is small relative to neu/pos. Charlotte OASIS science was not hashed the way Arnav eval is. Orthogonalizing to r does not make a refusal-independent [Wollschläger et al., 2025]. Publish ratio $_a$ used a repeated distress string. Detector AUROC 1.0 is measured against a white-box injection. Gemma-3-4B’s missing gate co-occurs with the lowest a_{stab} . Results are instruction-tuned Gemma and a small VLM roster, not a survey of deployed systems.

9 Broader impacts

Positive use: if the 12B gate is real, a runtime monitor on the affect projection can flag white-box steers, and natural emotional images do not appear to be an input-space jailbreak under the

tested protocols. Negative use: the same 12B result is a white-box exposure. A small coherent α on $a_{\perp}$ collapses refuse. We do not package an attack tool. Most scoring is generation-free. Arnav’s secondary Grok evaluation did generate short completions and send each one for refusal classification. The named JSON artifacts retain aggregate rates, not completion text. Image sets are existing public corpora, not newly scraped distress photos.

10 Conclusion

Emotional images did not lower refusal under the recorded protocols. On Gemma-3-12B, affect gates refusal through r when steered. On Gemma-4, the supported mechanism statement is MAGNITUDE_GAP plus a withdrawn INVERTED ranking, not a finished shared-appraisal theorem. The pipelines are unified in code and not yet unified in confirmatory numbers. This is a progress report.

Acknowledgments

Algoverse research program. Draft editing and pipeline-unification advising used LLMs. Grok was a secondary refusal judge. The experimental artifacts named above are the evidence for scientific claims.

References

- Andy Arditi, Oscar Obeso, Aaqib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems*, volume 37, 2024.
- Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibong Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, et al. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*, 2025.
- Gemma Team. Gemma 4 technical report, 2026. URL <https://arxiv.org/abs/2607.02770>.
- Gemma Team, Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, Sarah Perrin, Tatiana Matejovicova, Alexandre Ramé, Morgane Rivière, et al. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786*, 2025.
- Ronak Kosti, Jose M. Alvarez, Adria Recasens, and Agata Lapedriza. Emotion recognition in context. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 1667–1675, 2017.
- Ronak Kosti, Jose M. Alvarez, Adria Recasens, and Agata Lapedriza. Context based emotion recognition using EMOTIC dataset. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 42(11):2755–2766, 2019.
- Benedek Kurdi, Shayn Lozano, and Mahzarin R. Banaji. Introducing the open affective standardized image set (OASIS). *Behavior Research Methods*, 49(2):457–470, 2017.
- Bo Li, Yuanhan Zhang, Dong Guo, Renrui Zhang, Feng Li, Hao Zhang, Kaichen Zhang, Peiyuan Zhang, Yanwei Li, Ziwei Liu, and Chunyuan Li. LLaVA-OneVision: Easy visual task transfer. *Transactions on Machine Learning Research*, 2025. URL <https://mlanthology.org/tmlr/2025/li2025tmlr-llavaonevision/>.
- Neel Nanda and Joseph Bloom. TransformerLens. <https://github.com/TransformerLensOrg/TransformerLens>, 2022.
- Nina Rimskey, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15504–15522. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.828.
- Lihao Sun, Lewen Yan, Xiaoya Lu, Andrew Lee, Jie Zhang, and Jing Shao. Valence–arousal subspace in LLMs: Circular emotion geometry and multi-behavioral control. *arXiv preprint arXiv:2604.03147*, 2026.
- Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. Stanford Alpaca: An instruction-following LLaMA model. https://github.com/tatsu-lab/stanford_alpaca, 2023.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Steering language models with activation engineering. *arXiv preprint arXiv:2308.10248*, 2023.

Tom Wollschläger, Jannes Elstner, Simon Geisler, Vincent Cohen-Addad, Stephan Günnemann, and Johannes Gasteiger. The geometry of refusal in large language models: Concept cones and representational independence. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 66945–66970. PMLR, 2025. URL <https://proceedings.mlr.press/v267/wollschlager25a.html>.

Zhenhong Zhou, Haiyang Yu, Xinghua Zhang, Rongwu Xu, Fei Huang, and Yongbin Li. How alignment and jailbreak work: Explain LLM safety through intermediate hidden states. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 2461–2488, 2024.

Jinguo Zhu, Weiyun Wang, Zhe Chen, Zhaoyang Liu, Shenglong Ye, Lixin Gu, Hao Tian, Yuchen Duan, Weijie Su, Jie Shao, et al. InternVL3: Exploring advanced training and test-time recipes for open-source multimodal models. *arXiv preprint arXiv:2504.10479*, 2025.

Andy Zou, Zifan Wang, Nicholas Carlini, Milad Nasr, J. Zico Kolter, and Matt Fredrikson. Universal and transferable adversarial attacks on aligned language models. *arXiv preprint arXiv:2307.15043*, 2023.

A Reviewer questions

The main text already answers the load-bearing items (circularity, two stacks, $100\times$ versus ratio_a , nf4, leakage, Grok versus first-token). Table 7 compactly maps all fifteen lock-file questions.

Table 7: Answers to the lock-file reviewer questions. Pointers are to main-text sections.

Question	Answer
Why no image jailbreak if affect gates refusal?	The 12B stack records image movement $\sim 100\times$ below its steer ceiling. v3 separately records suppression 0.107 and $R_{\text{affect}} = 0.0050$ (Sec. 5.2, 5.4).
Is $\text{ratio}_a = 1.59$ evidence images $>$ text? Same method now?	No. Image-built axis, construction bias (Sec. 5.3). Shared CORE_METHOD in code; confirmatory rerun pending (Sec. 6).
Why two models and two corpora?	Generation, quantization, and image semantics are science factors (Table 2).
Is nf4 a confound versus bf16? Did train images leak into eval?	Yes, unquantified. Do not pool effect sizes (Sec. 8). Arnav: hashed splits. Charlotte OASIS: split-half, some in-sample estimates (Sec. 3).
Why first-token scores? When Grok?	Headline is generation-free. Grok is secondary on Arnav behavior: publish $n = 100$; v3 harmful $n = 100$, border $n = 80$. Publish fallback is 0% (Sec. 3).
Frozen versus rerun on unified core?	v3 JSON frozen. Core locked. Rerun not done (Sec. 6, 7).
Does orth to r make a independent?	No. 12B r -restore and Wollschläger (Sec. 2, 5.2).
Is $100\times$ the same as 0.107?	No. Reachability gap versus $\Delta_{\text{harm}}/\Delta_{\text{DESCRIBE}}$ (Sec. 2, 5.4).
Why Gemma-3-4B has no gate?	Affect-gate refuse 1.0 at $\alpha = 0.004$; lowest $a_{\text{stab}} = 0.633$ (Table 5).
What failed?	Sycophancy endpoint, color squares, EMOTIC 1-image poison, sequential orth, all-layer proj, $\text{coeff}=20$ dry run (Sec. 7).
Ethics: harmful completions?	Yes, short completions were generated for Arnav’s Grok judge. Named JSON artifacts retain aggregate rates, not completion text (Sec. 9).
Compute?	T4 publish ~ 8942 s; A100 v3 6740.5 s. Charlotte’s GPU and wall time are not recorded (Sec. 4).
Pre-registration?	Hashes in Sec. 4; three direction rebuilds not done.

B Additional recorded numbers

Publish γ table. Refuse stays 1.0 for $\gamma \in \{1, \dots, 20\}$. At $\gamma = 50$: emotion 0.95, random 1.0. At $\gamma = 100$: emotion 0.475, random 0.1. Coherent fraction 1.0 at every recorded γ . Interpretation: generic collapse.

Publish gap cells. $d_{\text{img},A} = 5.898$, $d_{\text{cap},B} = 6.660$, $d_{\text{txt},C} = 13.259$, $B/A = 1.129$, $n = 50$. Relative A versus B is usable as a caption $\approx$ photo check on the image-built axis. C is not a clean text-emotion upper bound.

v3 cosine table. $|\cos(a_{\text{text}}, r)| = 0.169$; $|\cos(a_{\text{img}}, r)| = 0.135$; $|\cos(a_{\text{text}}, a_{\text{img}})| = 0.098$; $|\cos(a_{\text{both}}, r)| = 0.186$; $|\cos(j, r)| = 0.247$.

Hashes. Publish prereg 3111c3af7f7b12656cb0792bea4a21b4302a0c42; publish split ef2a7d2a84a803c7; v3 split 1e8ea1c22144dd9d; sycophancy plan sha256 prefix 051e1ea79cd116fce56edeb43848ae10526e2830.

C Method variants that are out of scope

Obsolete Charlotte notebook row labeled “Arnav”: late layers, last-token only, `coeff=20`. Status: `obsolete_dry_run`. Current Arnav mechanism already used `normscaled + fractional` gate before unification.

All-layer mean projection is not the locked proj: it dilutes the gate window.

Sequential Gram-Schmidt against $\{r, j\}$ is not the locked orth.

Image-only $a_{\perp}$ may be reported as `image_only_a_perp`. It may not headline an image-versus-text ranking.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: Abstract and Section 1 tag each finding as HOLDS, WITHDRAWN, CODE LOCKED, or INCOMPLETE. We do not present the INVERTED ratio as a modality ranking, do not pool Gemma-4 nf4/EMOTIC with Gemma-3-12B bf16/OASIS effect sizes, and do not claim a confirmatory unified-core rerun.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 8 and Section 7 cover quantization, dual corpora, circularity of image-built axes, missing direction rebuilds, incomplete sycophancy, and the fact that unified-core code has not been re-run.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: This is an empirical progress report. Direction constructions and steering formulas are definitions, not theorems.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [No]

Justification: Sections 3 and 4 record the main method, models, split hashes, and Arnav compute. A full reproduction package, exact run commands and environments, model and dataset revisions, and complete Charlotte run provenance are not provided.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: This is an internal practice draft. Shared core code lives in the project repository and a unification pull request is recorded, but we do not package a public reproduction release with this writeup. EMOTIC, OASIS, AdvBench, and the Gemma checkpoints are existing public assets cited in Section 4.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 4 gives splits, hashes, ns , α , gate windows, quantization, prompts, and GPUs. There is no optimizer: directions are difference-in-means and interventions are forward-hook adds.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: Section 5.4 defines the 95% paired-image bootstrap CI $[0.079, 0.130]$ at $n_{\Delta} = 64$. The main behavioral rates and earlier geometry estimates lack intervals, and the planned three direction rebuilds were not run.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.).
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [No]

Justification: Section 4 records a Tesla T4 (~15.6 GB, 8942 s) for the Gemma-4 publish pipeline and an NVIDIA A100-SXM4-40GB (6740.5 s) for v3. Charlotte’s GPU SKU and wall time, failed-run compute totals, storage, and total project compute are not recorded.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn’t make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: Section 9 discloses the white-box steering exposure and that Arnav generated short completions for Grok classification. The named JSON artifacts retain aggregate rates rather than completion text; most other scoring is generation-free.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 9 states the defensive use (monitor the gate; images do not open it) and the negative path (white-box affect steering can collapse refusal on Gemma-3-12B).

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: We do not release a new model or scraped dataset. The writeup reports steering coefficients and layer windows on already public VLMs; that is discussed as exposure in Section 9, not as a packaged attack tool.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [No]

Justification: Section 2 and the bibliography credit the papers, datasets, software, and model reports. This internal draft does not yet name or audit the exact license and terms for each dataset, checkpoint, and software version.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset’s creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [N/A]

Justification: This practice writeup does not release a new dataset, model, or public code package. Frozen JSON artifacts already exist inside the project and are not modified here.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: No new human-subjects or crowdsourcing study. OASIS and EMOTIC are existing published image sets; we did not re-collect ratings.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: No human-subjects protocol. Image labels come from published datasets.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does not impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: LLMs assisted draft editing and pipeline-unification advice. They are not the method under test. Grok classified short individual generations as a secondary judge; the paper reports aggregate rates, with first-token scores as the generation-free headline.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.